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Sreejith Institute of Medical And Technical Sciences
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CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	D.S.Tharuniya Priya	Register Number	192511317
Class / Section	A	Slot	B
Course Code	CSA1512	Course Title	Cloud computing and Big Data Analytics
Capstone Title	PulsePay – Cloud fake Payment Screenshot Detector with Merchant Verification Workflow		
Submission Date	13 July 2026		
Faculty Name	Dr.Rajesh Babu	Employee ID	
Department			

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	D.S.Tharuniya Priya
Date	13 July 2026

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

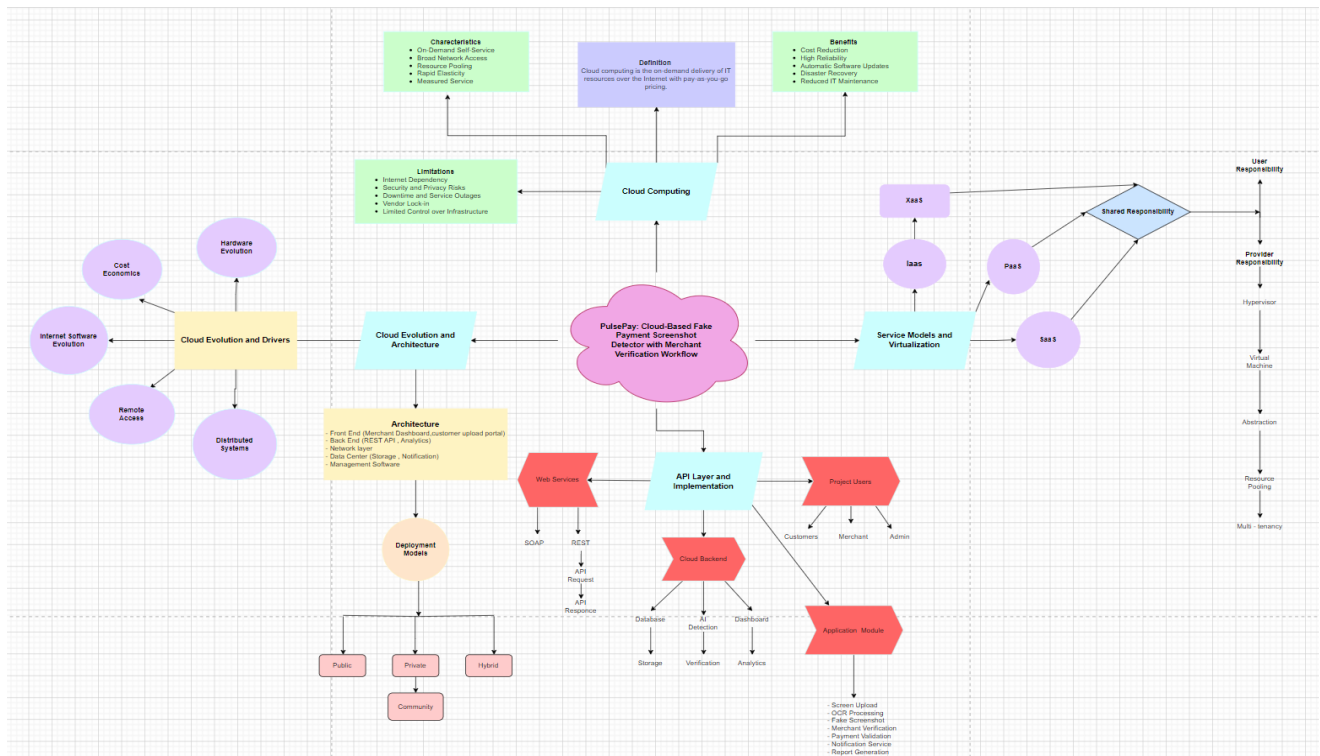
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

The concept map centers on the PulsePay capstone project and shows how cloud computing supports fake payment screenshot detection. It includes key cloud concepts such as characteristics, service models, deployment models, security, storage, and architecture. The labeled links (e.g., uses, provides, communicates through, stores in) illustrate the relationships between components. The hierarchy progresses from cloud computing fundamentals to application modules and system architecture, demonstrating how cloud services enable secure, scalable, and efficient payment verification.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Virtual servers & storage VM/Cloud setup screenshot	Manage servers, networking, storage	Configure VM and deploy application	VM/Cloud setup screenshot
PaaS	Backend API & application hosting	Manage runtime, OS, middleware	Develop and deploy backend code	Backend deployment screenshot
SaaS	Merchant & Customer Web/App	Provide application availability	Manage users and application features	Application screenshots
XaaS / Specialized Service	OCR & Notification Service	Provide OCR API and notification service	Integrate APIs into application	API integration screenshot

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	Client access and UI	Client access and UI	Client access and UI	Client access and UI
API / backend logic	Android App & Web Dashboard	Android App & Web Dashboard	Android App & Web Dashboard	Android App & Web Dashboard
Authentication and identity	User inputs & screenshots	User inputs & screenshots	User inputs & screenshots	User inputs & screenshots
Database / structured records	HTTPS REST API	HTTPS REST API	HTTPS REST API	HTTPS REST API
File / object storage	Secure login	Secure login	Secure login	Secure login
Monitoring / logs / usage metrics	API / Backend logic	API / Backend logic	API / Backend logic	API / Backend logic
Analytics / reporting output	REST API (Node.js/Spring Boot)	REST API (Node.js/Spring Boot)	REST API (Node.js/Spring Boot)	REST API (Node.js/Spring Boot)

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

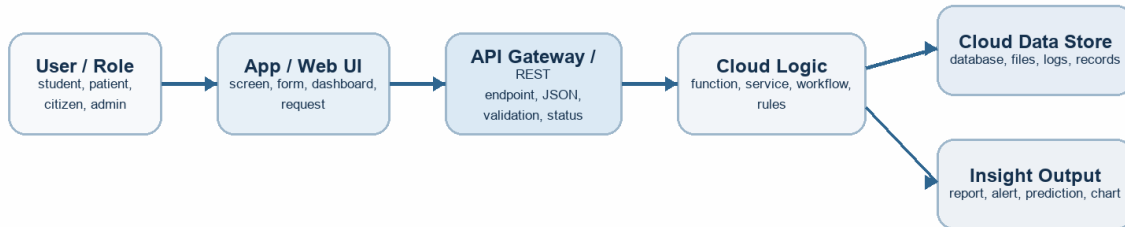
The selected cloud service pattern is suitable for the PulsePay capstone because it provides secure user authentication, cloud storage for payment screenshots, and a scalable backend for processing OCR and verification requests. REST APIs enable smooth communication between the mobile app, web dashboard, database, and cloud services. Cloud infrastructure supports increasing numbers of users without affecting performance, while encryption, authentication, and secure storage protect sensitive payment and merchant data. This architecture ensures reliable, efficient, and secure fake payment screenshot detection.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

Sample Capstone Cloud Flow Template

Replicate this structure using your own capstone screens, APIs, cloud services, data stores, and outputs.



Cross-cutting cloud decisions to label in your own diagram

Authentication / IAM

Who is allowed to access each API and data item?

Scalability

Which component must handle growing users or records?

Elasticity

Which component can expand or shrink based on demand?

Monitoring

What logs, usage metrics, or errors will be observed?

Security

Where are validation, privacy, backup, and access control applied?

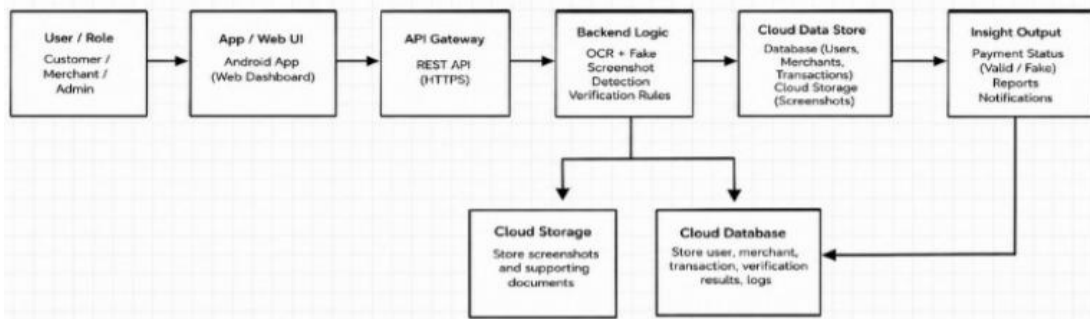
Student instruction: redraw this flow in draw.io/diagrams.net and replace every label with your own capstone architecture.

Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	Customer uploads payment screenshot; Merchant verifies payment; Admin monitors reports.
Frontend / app screen	Android App (Upload Screen) and Merchant Web Dashboard.
API endpoint and method	POST /api/upload, POST /api/verify, GET /api/report (REST API).
Backend processing rule	OCR extracts text → AI checks for fake screenshot → Merchant details verified → Payment status generated.
Cloud database / storage operation	Store screenshots in Cloud Storage; Save user, merchant, transaction, and verification records in Cloud Database.
Response, alert, report, or dashboard output	Display Valid/Fake Payment result, send notification, and generate fraud reports on the dashboard.
Security, scalability, and elasticity labels added in diagram	Authentication, HTTPS Encryption, Role-Based Access Control, Auto Scaling, Cloud Storage, Load Balancing, Monitoring.

Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

A customer uploads a payment screenshot through the Android app. The app sends the image to the backend via a secure REST API, where OCR extracts the payment details and the system verifies the transaction with the merchant. The screenshot and verification results are stored in the cloud database. Finally, the cloud returns the payment status (Valid or Fake) to the user and updates the merchant dashboard with notifications and reports.

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud concept is cloud-based scalability and secure storage, which enable real-time fake payment screenshot detection and merchant verification for multiple users.

2. Where will elasticity or scalability become important in your capstone product?

Elasticity and scalability are important when many users upload payment screenshots simultaneously. Cloud resources can automatically scale to maintain fast processing and system performance.

3. Which API or web-service idea is most relevant to your project and why?

A REST API is the most relevant because it enables secure communication between the mobile app, web dashboard, backend, OCR service, and cloud database for payment verification.

4. What did you personally understand better after preparing this concept map and matrix?

I understood how cloud services, APIs, storage, security, and scalability work together to build a complete cloud-based application and support efficient data processing.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	